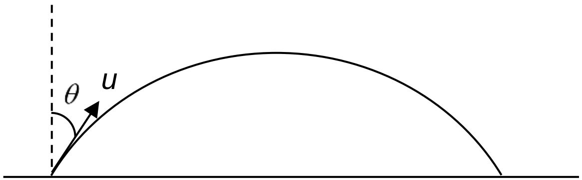
10 The diagram shows a projectile being launched at an angle θ to the vertical.



Ignoring air resistance, what fraction of its initial kinetic energy does the projectile have at the top of its trajectory?

A	zero	**B**	$\cos\theta$	**C**	$\cos^2\theta$	**D**	$\sin^2\theta$

L2 **Answer: D**

Initial kinetic energy = $\frac{1}{2}mu^2$

At the top of its trajectory, its velocity is $u\sin\theta$,

Therefore, its kinetic energy is $\frac{1}{2}m(u\sin\theta)^2$

Fraction = $\dfrac{\textit{kinetic energy at maximum height}}{\textit{initial kinetic energy}} = \sin^2\theta$

11 An object of mass m moves in a circular path of radius r at a constant angular speed ω